

PART-B

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Signature of the Invigilator: Agnily

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	0	6	2	1	09

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yx} \\ S_{yx} & S_{xx} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

Given $Y_{1 \times 1}$; $X_{q \times 1}$ sample size n , centered wrt mean.

first canonical correlation $r_1^2 \rightarrow$ between Y & X

Y being 1×1 cannot be projected $\therefore$ we wish to maximize $\text{Corr}(a^T X, Y)$ maximize. $\rightarrow$ find such an a

It is maximized at the eigen vector e_1 of $S_{xx}^{-1/2} S_{xy} S_{yy}^{-1} S_{yx} S_{xx}^{-1/2}$ and the maximum

correlation attained is at the corresponding λ_1 eigen value.

~~Proof~~

Regression finds the combination $\beta^T X$ s.t. maximum amount of variance in Y is explained by $\hat{\beta}^T X$ while minimizing the error.

for that to happen we need $\text{Corr}(\hat{\beta}^T X, Y)$ to be as much as possible & as regression is BLUE it achieves same $\hat{\beta}^T = a^T$ as canonical correlation analysis.

if β were to be standardized like a then we.

arrive at the exact conclusion that the first/maximum canonical correlation b/w Y, X given by r_1^2 is what is achieved by the coefficient of determination (R^2) while regressing,

↳ otherwise $\hat{\beta}$ cannot be BLUE.

$$\therefore r_1^2 = R^2$$

GO

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix S . The eigenvalues of S are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are e_1, e_2, e_3 , and e_4 . You decide to extract $m = 2$ factors using the principal component method.
- (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{L}$ in terms of the given eigenvalues and eigenvectors.
 - (1 point) Let the first row of $\tilde{L}$ be $\tilde{l}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
 - (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of S .

a) The factor loadings

$$L = [\sqrt{\lambda_{(1)}} e_{(1)}, \sqrt{\lambda_{(2)}} e_{(2)}]$$

Largest to smallest
rearrangement.

$$\lambda_{(1)} = 4.2, \lambda_{(2)} = 2.1, \lambda_{(3)} = 0.8, \lambda_{(4)} = 0.2$$

what are $e_{(i)}$??

b) first row $[0.8, -0.4]$ $\rightarrow$ estimated communality $(h_1^2) = \lambda_{11}^2 + \lambda_{12}^2$

$$\Rightarrow 0.8^2 + (-0.4)^2 = 0.64 + 0.16 = 0.8$$

c) The sum of all estimated communalities

$$\sum_{i=1}^p \tilde{h}_i^2 = \sum_{i=1}^p \sum_{j=1}^m \tilde{\lambda}_{ij}^2$$

$$= \sum_{j=1}^m \sum_{i=1}^p \tilde{\lambda}_{ij}^2$$

$$= \sum_{j=1}^m \lambda_{(j)}$$

$$= \lambda_1 + \lambda_2 + \dots + \lambda_m$$

$$\tilde{\lambda}_{(i)j} = \sqrt{\lambda_{(j)}} e_{(j)}$$

$$\sum_{i=1}^p \tilde{\lambda}_{ij}^2 = \|\tilde{\lambda}_{(j)}\|^2$$

$$= \|\sqrt{\lambda_{(j)}} e_{(j)}\|^2$$

$$= (\sqrt{\lambda_{(j)}})^2 \|e_{(j)}\|^2$$

$$= \lambda_{(j)} \quad \downarrow \text{normalized hence} = 1$$

But m was chosen to be 2

$$\therefore \sum_{i=1}^p \tilde{h}_i^2 = \underbrace{\lambda_{(1)} + \lambda_{(2)}}_{\text{two largest eigenvalues}}$$

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function.

Prove that f is differentiable at x_0 if and only if

$$\lim_{h \rightarrow 0} \frac{\|f(x_0 + h) - f(x_0) - Df(x_0)h\|}{\|h\|} = 0$$

where $Df(x_0)$ is the Jacobian matrix of f at x_0 .

$$Df(x_0) = (Df_1(x_0), \dots, Df_m(x_0))$$

$$Df_i(x_0) = \left(\frac{\partial f_i}{\partial x_1}(x_0), \dots, \frac{\partial f_i}{\partial x_n}(x_0) \right)$$

for this to be independent of n we have

$$Df(x_0) = \begin{pmatrix} Df_1(x_0) \\ \vdots \\ Df_m(x_0) \end{pmatrix} \in \mathbb{R}^{m \times n}$$

$$\frac{\partial f_i}{\partial x_j}(x_0) = \lim_{h \rightarrow 0} \frac{f_i(x_0 + h e_j) - f_i(x_0)}{\|h\|}$$

$$\frac{\partial f_i}{\partial x_j}(x_0) = \lim_{h \rightarrow 0} \frac{f_i(x_0 + h e_j) - f_i(x_0)}{\|h\|} = \lim_{h \rightarrow 0} \frac{f_i(x_0 + h e_j) - f_i(x_0)}{h}$$

$$\frac{\partial f_i}{\partial x_j}(x_0) = \lim_{h \rightarrow 0} \frac{f_i(x_0 + h e_j) - f_i(x_0)}{h} = \lim_{h \rightarrow 0} \frac{f_i(x_0 + h e_j) - f_i(x_0)}{h}$$

if we let $f_i(x) = x^2$ then we have $\frac{\partial f_i}{\partial x_j}(x_0) = 2x_0$ for all j and i .

if $f_i(x) = x^2$ then we have $\frac{\partial f_i}{\partial x_j}(x_0) = 2x_0$ for all j and i .

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

$X_1, \dots, X_n$ R.S. mean μ , variance μ^2

↳ Variance stabilizing transformation $g(\bar{X}_n)$ of $\bar{X}_n$

$$\text{Var}(X_i) = \mu^2 \rightarrow \text{Var}(\bar{X}_n) = \frac{\mu^2}{n} = \sigma_{\bar{X}_n}^2$$

from Δ method & Taylor's expansion we know that

$$g(\bar{X}_n) - g(\mu) \approx g'(\mu) (\bar{X}_n - \mu)$$

$$\text{Var}(g(\bar{X}_n)) \approx (g'(\mu))^2 \times \sigma_{\bar{X}_n}^2 = \frac{\mu^2}{n} (g'(\mu))^2$$

for this to be independent of μ we have.

$$(g'(\mu))^2 \times \frac{\mu^2}{n} = C^2 \rightarrow \text{some constant} \\ \text{let 1 for now}$$

$$g'(\mu) = \frac{\sqrt{n}}{\mu}$$

~~$$g'(x) = \frac{1}{x} \text{ for any } x \in \mathbb{R}$$~~

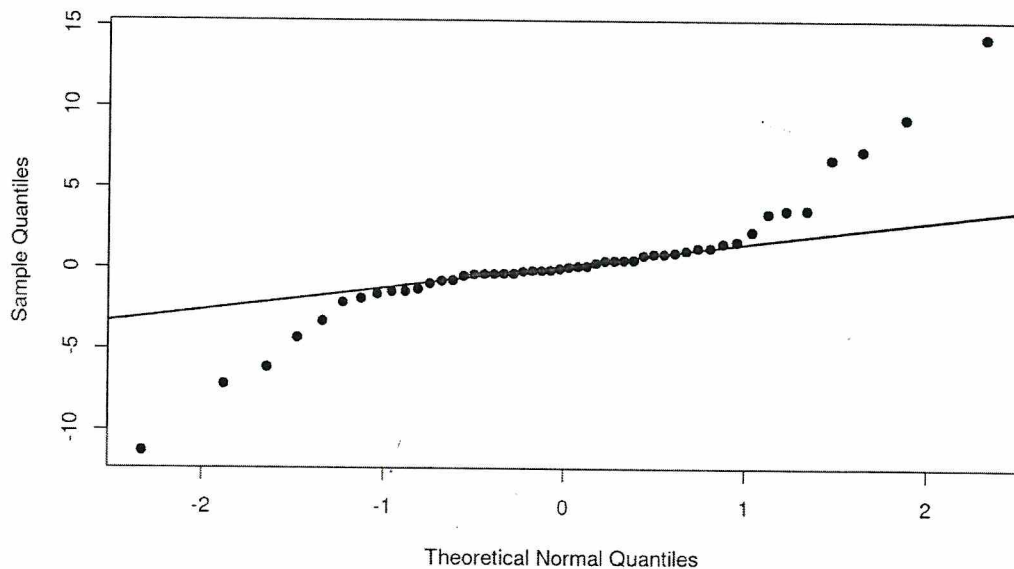
$$g'(x) = \frac{\sqrt{n} x}{\mu^2} \quad \text{or} \quad g(x) = \int \frac{\sqrt{n} x}{\mu^2} dx = \frac{\sqrt{n} x^2}{2\mu^2} + C'$$

02 $g(x) = \frac{\sqrt{n} x^2}{2\mu^2}$ gives us unit Variance for $g(\bar{X}_n)$

if $\mu \in \mathbb{R}^+$ however we could set $g'(x) = \frac{\sqrt{n}}{x}$ to obtain $\sqrt{n} \log x$ as the variance stabilizing transformation.

if $\mu \in \mathbb{R} \setminus \{0\}$ then $\sqrt{n} \log |x|$ would also work.

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



Given the above Q-Q plot, The data doesn't follow Normal distribution as it is not approximately on the line $y=x$

↳ there is a significant but almost symmetric deviation at both ends of the plot.

This graph indicates that at the tails, the probability for a point to be in $(-2, -1)$ (or) $(1, 2)$ when sampled from standard normal is same as probability for point to be in $\approx (-10, -1)$ (or) $(1, 10)$ when observed from sample.

↳ The class of distributions correspond to thin ~~tailed~~ distributions which have kurtosis < 3 ;

~~heavy tailed distributions which have kurtosis > 3 ;~~

01

ADDITIONAL PAGES FOR ANSWERS

Q. 1. (a) A particle of mass m is moving in a circular path of radius r with a constant speed v . Find the work done by the centripetal force in one complete revolution.

(b) A particle of mass m is moving in a circular path of radius r with a constant speed v . Find the change in momentum in one complete revolution.

(c) A particle of mass m is moving in a circular path of radius r with a constant speed v . Find the change in kinetic energy in one complete revolution.

(d) A particle of mass m is moving in a circular path of radius r with a constant speed v . Find the change in potential energy in one complete revolution.

(e) A particle of mass m is moving in a circular path of radius r with a constant speed v . Find the change in total energy in one complete revolution.

